AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 13 · STUDENT LESSON

Effect size and practical importance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Uncertainty intervals. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of effect size and practical importance. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **effect size** and practical importance.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **effect size** and practical importance

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Uncertainty intervals. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **effect size** and practical importance using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Effect size and practical importance case. Separate observations, given facts and assumptions before choosing an explanation.

Effect size describes magnitude on a stated scale. Practical importance depends on task requirements, not merely a nonzero difference.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Effect size and practical importance. Connect each labelled input, deciding step and result.

For paired response durations compute baseline minus candidate per case and average the differences. Positive means faster candidate. Compare with a prespecified minimum useful gain.

Effect size and practical importance — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Baseline times10, 12, 8; candidate8, 9, 7 seconds. Useful mean gain requires at least 2 seconds.	For paired response durations compute baseline minus candidate per case and average the differences. Positive means faster candidate. Compare with a prespecified minimum useful gain.	Paired gains2, 3, 1; mean2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Effect size and practical importance. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Baseline times10, 12, 8; candidate8, 9, 7 seconds. Useful mean gain requires at least 2 seconds. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: B makes 8 errors in 1000.

Predicted result

Prediction: The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

Why the result follows

Reason: Twenty minus eight exceeds the required ten fewer errors.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Effect size and practical importance to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Effect size	The magnitude of a difference under a declared scale
Paired difference	A within-unit comparison
Practical threshold	A prespecified magnitude considered useful

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Effect size and practical importance, predict a step, then compare with its explanation.

Baseline times10, 12, 8; candidate8, 9, 7 seconds. Useful mean gain requires at least 2 seconds.

Paired gains2, 3, 1; mean2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty.

Worked explanation · Effect size and practical importance

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Worked explanation · Effect size and practical importance

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def paired_gain(baseline, candidate):
    if not baseline or len(baseline) != len(candidate):
        raise ValueError('paired lengths')
    differences = [a - b for (a, b) in zip(baseline, candidate)]
    return (differences, sum(differences) / len(differences))

print(paired_gain([10, 12, 8], [8, 9, 7]))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Effect size and practical importance

An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.

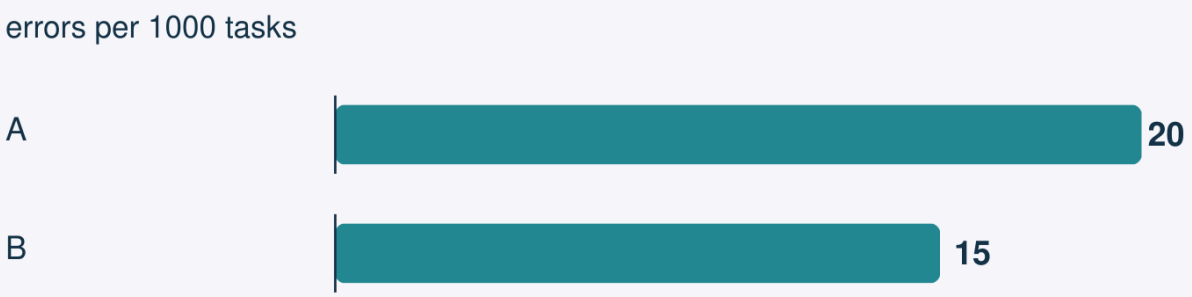
The facts we will use

Supplied facts

A makes 20 errors in 1000 tasks; B makes 15. Practical improvement requires at least 10 fewer errors per 1000.

AI ACADEMY / AI-7/13

Errors on 1000 tasks



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/13

Effect size and practical importance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute rates

A is 2%; B is 1.5%.



2. Compute effect size

B has 5 fewer errors per 1000, a 0.5-percentage-point reduction.



3. Apply practical threshold

Five is less than ten, so the observed change misses this threshold.

CHANGE ONE THING

B makes 8 errors in 1000.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute rates	A is 2%; B is 1.5%.
2. Compute effect size	B has 5 fewer errors per 1000, a 0.5-percentage-point reduction.
3. Apply practical threshold	Five is less than ten, so the observed change misses this threshold.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Effect size describes how much the outcome changes. Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

What this example does not establish

Keep the stated limit

No uncertainty or statistical significance is supplied; these counts alone cannot establish a general effect.

Keep the stated limit

Read the labelled picture: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

Keep the stated limit

Change one thing: B makes 8 errors in 1000. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Effect size and practical importance claim and explain a repair using the supplied evidence.

A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Effect size and practical importance with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Effect size** and practical importance". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Effect size** and practical importance. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Effect size and practical importance. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Baseline times10, 12, 8; candidate8, 9, 7 seconds. Useful mean gain requires at least 2 seconds. Show the intermediate steps and give the conclusion with its scope.

```
def paired_gain(baseline, candidate):
    if not baseline or len(baseline) != len(candidate):
        raise ValueError('paired lengths')
    differences = [a - b for (a, b) in zip(baseline, candidate)]
    return (differences, sum(differences) / len(differences))

print(paired_gain([10, 12, 8], [8, 9, 7]))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Effect size and practical importance. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled **effect size** and practical importance **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **effect size** and practical importance.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Effect size and practical importance case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Effect size and practical importance — evidence trace

INPUT AND SCOPE

Baseline times10, 12, 8; candidate8, 9, 7 seconds. Useful mean gain requires at least 2 seconds.

CHECK OR CALCULATION

For paired response durations compute baseline minus candidate per case and average the differences. Positive means faster candidate. Compare with a prespecified minimum useful gain.

CONCLUSION AND LIMIT

Paired gains2, 3, 1; mean2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Baseline times are 6, 8, 10 seconds; candidate times are 5, 7, 9 seconds on the same three cases in the same order. Define each gain as baseline minus candidate. Useful mean gain must be at least 1.5 seconds. Calculate and decide.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Baseline times are 6, 8, 10 seconds; candidate times are 5, 7, 9 seconds on the same three cases in the same order. Define each gain as baseline minus candidate. Useful mean gain must be at least 1.5 seconds. Calculate and decide.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Effect size and practical importance, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.

By the end, I can	How I will show it
Explain the mechanism and terminology of effect size and practical importance	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Effect size and practical importance without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **effect size** and practical importance

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Five is less than ten, so the observed change misses this threshold.

1. Apply **practical threshold**

2. Compute rates

3. Compute **effect size**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply **practical threshold** Five is less than ten, so the observed change misses this threshold. **Effect size** describes how much the outcome changes.

Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: B makes 8 errors in 1000. Which conclusion is supported by the supplied facts?

1. The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The reduction becomes 12 per 1000 and meets the stated **practical threshold**. Twenty minus eight exceeds the required ten fewer errors.

Repair a missing building block

Review Uncertainty intervals (ai-7/12). Goal: Explain the **mechanism** and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the **mechanism** and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

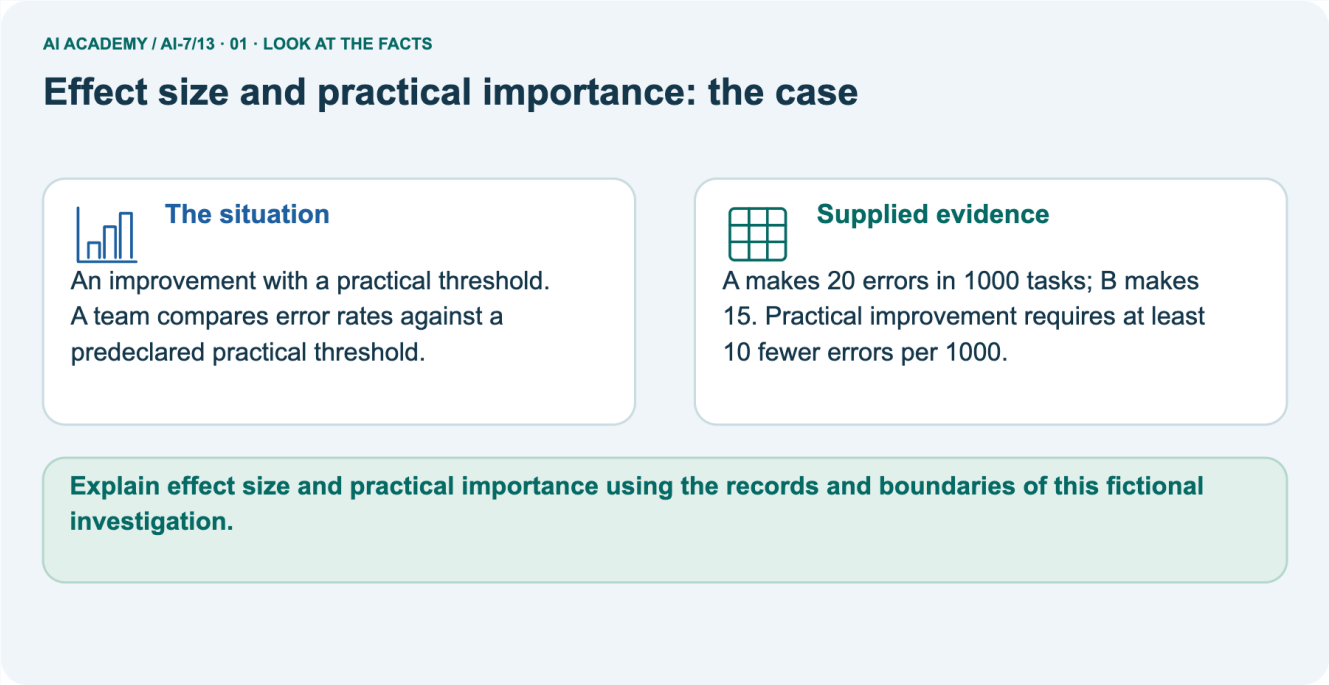
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Effect size and practical importance graphic using its labels, deciding evidence and relevant limitation.

01. Effect size and practical importance: the case



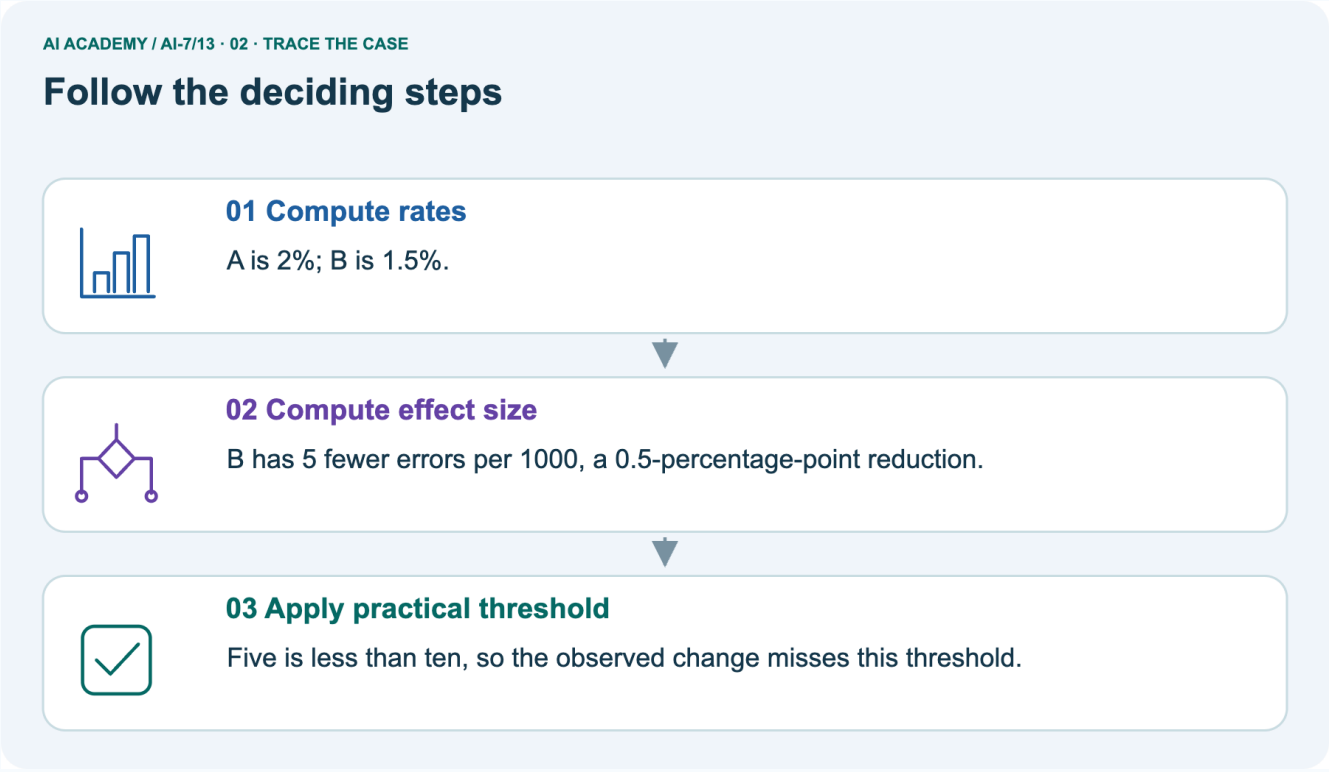
Follow the labelled picture with the matching explanation. Open the vector diagram

An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**. Supplied facts: A makes 20 errors in 1000 tasks; B makes 15.

Practical improvement requires at least 10 fewer errors per 1000.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute rates: A is 2%; B is 1.5%. → Compute **effect size**: B has 5 fewer errors per 1000, a 0.5-percentage-point reduction. → Apply **practical threshold**: Five is less than ten, so the observed change misses this threshold.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Errors on 1000 tasks

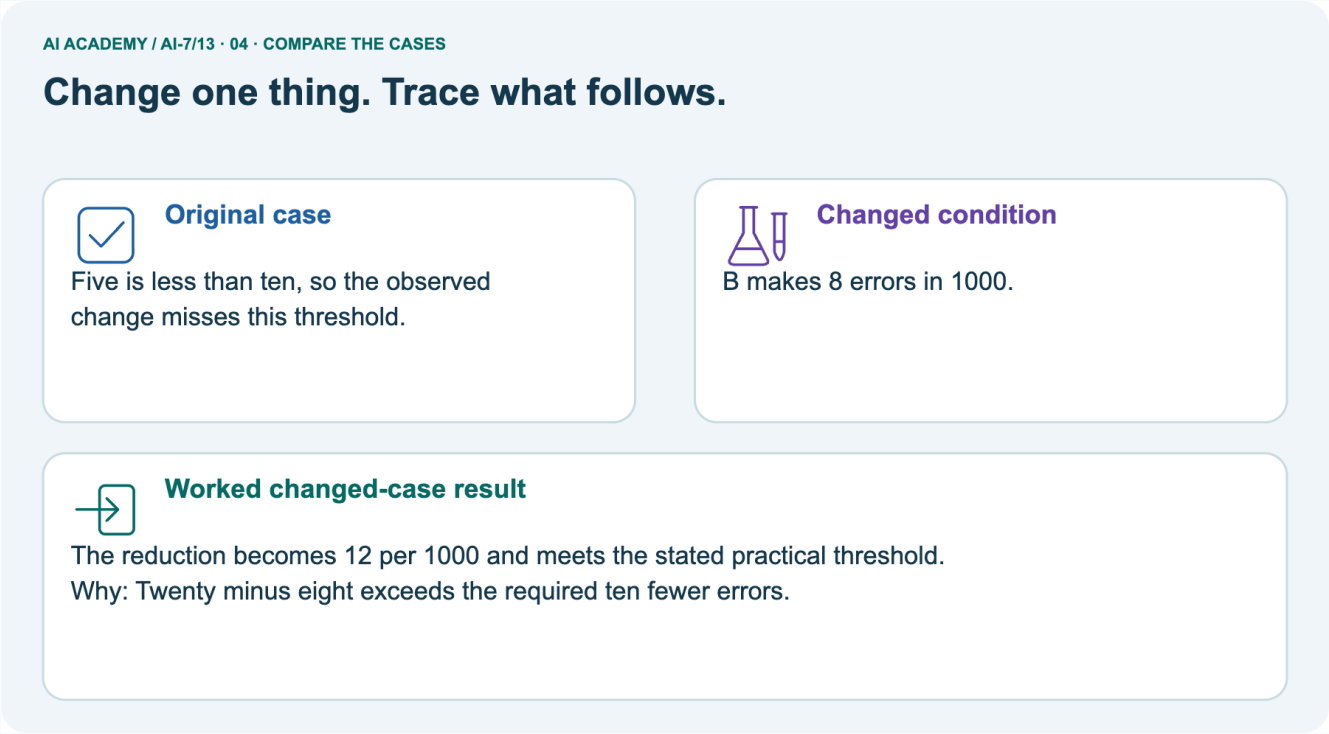


Follow the labelled picture with the matching explanation. Open the vector diagram

Errors on 1000 tasks. Units: errors per 1000 tasks. A: 20. B: 15.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



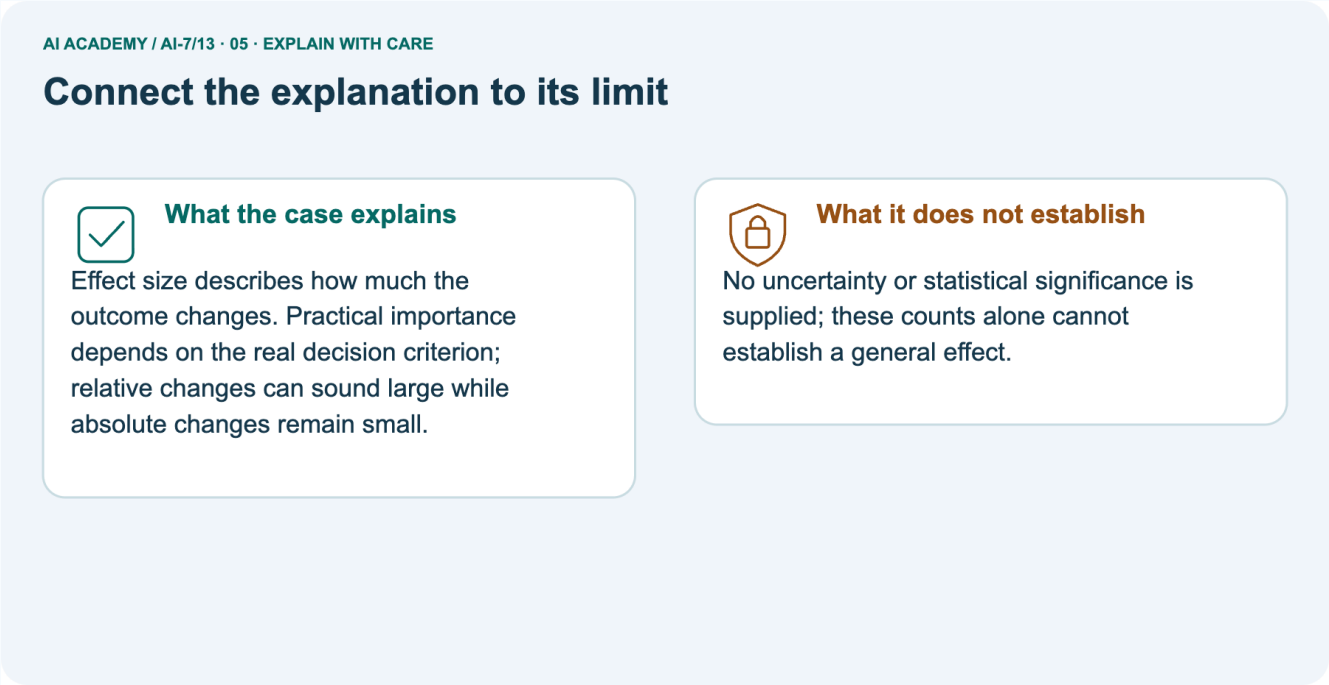
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Five is less than ten, so the observed change misses this threshold. Changed condition: B makes 8 errors in 1000. Expected result: The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

Reason: Twenty minus eight exceeds the required ten fewer errors.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



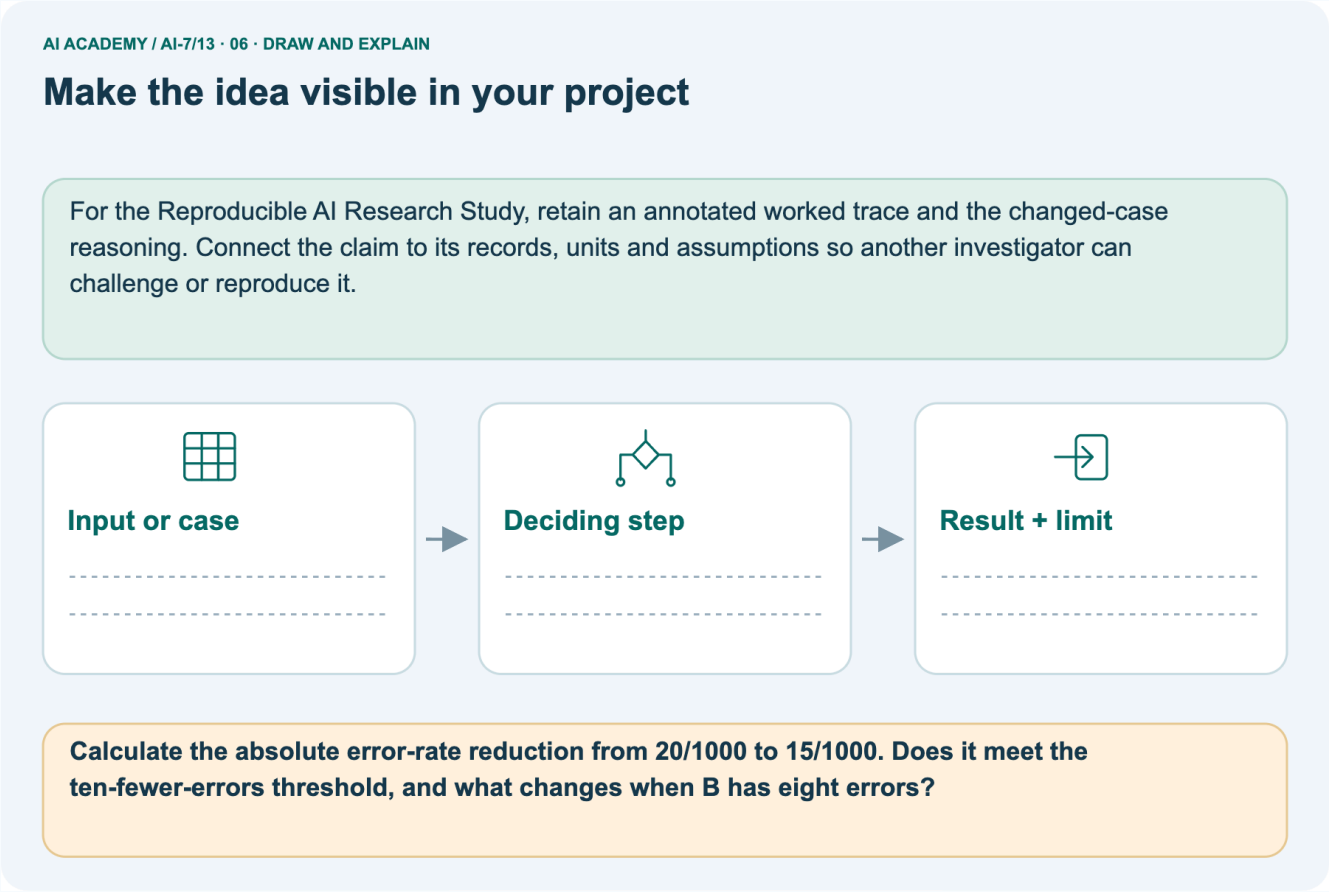
Follow the labelled picture with the matching explanation. Open the vector diagram

Effect size describes how much the outcome changes. Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

Limit: No uncertainty or statistical significance is supplied; these counts alone cannot establish a general effect.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.